

Flexible Combined Heat and Power Dispatch Based on Probabilistic Prediction of Wind Power

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Abstract—Economic development is accompanied by an increase in energy demand, and the energy crisis and environmental pollution problems. The integrated energy system can utilize the coordinated and complementary characteristics of multiple energy sources and the heterogeneous dynamic characteristics of multiple energy flows, which can promote the development of clean energy, intelligent production, and efficient energy consumption. A combined thermoelectric scheduling method based on probabilistic prediction of wind power generation under different control modes of thermal systems is proposed in this paper. First, extreme learning machine method is used as the probabilistic prediction model of wind power generation. On this basis, the combined heat and power scheduling model considering different control modes is established. Finally, a test system of an electric-thermal integrated energy system is taken as an example. The results show that considering the uncertainty of wind power prediction, the control method has the optimality.

Keywords—combined heat and power dispatch, flexibility, probabilistic prediction

I. INTRODUCTION

In order to cope with fossil fuels depletion and the environment pollution, China plans to construct new power system with new energy as the main part. By the end of 2021, The installed capacity of wind power and photovoltaic in China reaches 328GW and 306GW, respectively, and the access of a high proportion of renewable energy becomes an important factor affecting the flexible and safe operation of the power system. Regional integrated electric-heat energy system, as an important carrier of energy Internet, can effectively utilize the flexible adjustment characteristics of the heat system through electric-heat coupling equipment to realize the synergy and complementary of the electric heating system, improve the consumption level of renewable energy, and provide strong support for the implementation and accelerated development of energy transformation.

Accurate and reliable prediction can predict the change trend of renewable energy in advance, which is one of the key technologies to promote the absorption of new energy and provide key support for the safe and economic operation of power system [1]-[2]. A novel deeply asymmetric Laplacian neural network wind power prediction model is proposed in [3], and the asymmetric Laplacian distribution is used to characterize the uncertainty of wind power prediction. In order to construct high-quality forecast intervals and accurately capture uncertainties of wind power, a short-term wind power interval prediction method was proposed in [4]. An integrated time-domain volume integral digit regression network is adopted in [5] to avoid iterative propagation and gradient disappearance, and very long sequences can be processed in parallel. A quantile regression forest model for power interval prediction of ultra-short-term wind power was established in [6], which included power fluctuation process. Deep learning and neural networks have been widely used in wind power prediction.

In recent years, the electric-thermal integrated energy system has developed rapidly and attracted wide attention [7]-[8]. It can significantly improve energy efficiency, increase operational flexibility and facilitate the inclusion of renewable energy sources. A thermal power station system that can generate heat and meet heating requirements is proposed in [9], and the dynamic characteristics of electrical components and heating components are analyzed. A new furnace - machine - thermal cooperative control method is proposed in [10] to improve the operating flexibility of the heating system. Considering the power demand elasticity of the load, [11] took the hydrogen storage active load into the system to analyze its optimal operation mechanism. An optimal management system structure of layered predictive control is proposed in [12], and thermal storage tanks are used to supply energy to the district heating network. These studies have carried out a series of work on combined heat and power dispatching. However, with the continuous increase of wind power penetration, it is necessary to consider the uncertainty and develop scheduling

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models under different control modes. This paper introduces a method of combined heat and power scheduling based on probabilistic prediction of wind power under different control modes of thermal systems.

The rest of this paper is described as follows: The probabilistic prediction model for wind power generation is described in Section 2. In the third section, the combined thermoelectric scheduling model under different control modes is constructed. Section 4 presents a numerical study. Finally, a conclusion is drawn in section 5.

II. PROBABILISTIC PREDICTION OF WIND POWER USING EXTREME LEARNING MACHINE

Due to the non-stationarity of wind power sequence, the accuracy of traditional point prediction is not high, which increases the uncertainty and risk of system operation. Therefore, a probabilistic prediction method of wind power generation based on extreme Learning Machine (ELM) is adopted in this paper. ELM algorithm has the advantages of fast computing speed, simple structure and strong generalization ability. It solves many problems faced by traditional gradient descent algorithm. It can be modeled as

$$\sum_{j=1}^N \beta_j g(w_i \cdot x_j + b_i) = y_i, j = 1, 2, \dots, N \quad (1)$$

where N is the number of training set $\{(x_i, y_i) \mid x_i \in R^n, y_i \in R^m, i = 1, \dots, N\}$. The number of neurons in the input layer, hidden layer and output layer is n , h and m , respectively. $g(x)$ is the activation function. β_i is the weight of the i th hidden layer and b_i is the threshold. w_i is the weight of the i th input layer.

The calculation process of ELM does not require iteration, and w_i and b_i are randomly initialized. The loss function of the extreme learning machine is set as:

$$E(\theta) = |\beta - T| \quad (2)$$

where $\theta = \{\omega_i, b_i, \beta_i\}$; H is the output matrix of hidden layer, and T is the output expectation matrix. The problem of training ELM can be converted to solving this linear equation

$$\beta = H^{-1}T \quad (3)$$

where H^{-1} is the Moore–Penrose generalized inverse of matrix H .

Considering the noise caused by model uncertainty, it is necessary to establish a probabilistic prediction model of wind power generation. We assume that noise obeys Gaussian distribution and the variance $\sigma_\varepsilon^2(x_i)$ depends on the input vector

$$\varepsilon(x_i) \sim N_G(0, \sigma_\varepsilon^2(x_i)) \quad (4)$$

Following the bootstrap procedures, the model uncertainty intervals of ELM forecasts $I_M^\alpha(x_i) = [L_M^\alpha(x_i), U_M^\alpha(x_i)]$ can be obtained through

$$L_M^\alpha(x_i) = \hat{y}(x_i) - z_{1-\alpha/2} \sqrt{\sigma_{\hat{y}}^2(x_i)} \quad (5)$$

$$U_M^\alpha(x_i) = \hat{y}(x_i) + z_{1-\alpha/2} \sqrt{\sigma_{\hat{y}}^2(x_i)} \quad (6)$$

where $\hat{y}(x_i)$ is the trained neural network, which can be regarded as an estimation of the true regression $y(x_i)$; $z_{1-\alpha/2}$ is the critical value of the standard Gaussian distribution, which depends on the confidence level $100(1 - \alpha)\%$.

III. MODEL OF COMBINED HEAT AND POWER DISPATCH UNDER DIFFERENT CONTROL MODES

The combined heat and power dispatching model includes four control modes: thermoelectric unit, power grid, thermal network and thermal system.

A. Model of Heat and Electric Generation Units

In the electrothermal integration system, the generating units mainly include cogeneration units (TGU), cogeneration units (CHPs) and wind turbines (WTs). Cogeneration can generate electricity and heat through cascades of energy. gb is used as a heat source for peak regulation and an auxiliary heat source when the heat load peaks. The relevant operation constraints of power generation and heating units are as follows

$$P_{g,t}^{\text{TGU}} = \eta_g^{\text{TGU}} F_{g,t}^{\text{TGU}} \quad (7)$$

$$P_{c,t}^{\text{CHP}} = \xi_c^{\text{CHP}} H_{c,t}^{\text{CHP}}, P_{c,t}^{\text{CHP}} + H_{c,t}^{\text{CHP}} = \eta_c^{\text{CHP}} F_{c,t}^{\text{CHP}} \quad (8)$$

$$H_{b,t}^{\text{GB}} = \eta_b^{\text{GB}} F_{b,t}^{\text{GB}} \quad (9)$$

$$\underline{P}_c^{\text{CHP}} \leq P_{c,t}^{\text{CHP}} \leq \overline{P}_c^{\text{CHP}}, \underline{Q}_c^{\text{CHP}} \leq Q_{c,t}^{\text{CHP}} \leq \overline{Q}_c^{\text{CHP}} \quad (10)$$

$$\underline{H}_c^{\text{CHP}} \leq H_{c,t}^{\text{CHP}} \leq \overline{H}_c^{\text{CHP}} \quad (11)$$

$$\underline{H}_b^{\text{GB}} \leq H_{b,t}^{\text{GB}} \leq \overline{H}_b^{\text{GB}} \quad (12)$$

$$|P_{g,t}^{\text{TGU}} - P_{g,t-1}^{\text{TGU}}| \leq \Delta \overline{P}_g^{\text{TGU}} \quad (13)$$

$$|P_{c,t}^{\text{CHP}} - P_{c,t-1}^{\text{CHP}}| \leq \Delta \overline{P}_c^{\text{CHP}} \quad (14)$$

$$|H_{b,t}^{\text{GB}} - H_{b,t-1}^{\text{GB}}| \leq \Delta \overline{H}_b^{\text{GB}} \quad (15)$$

where $P_{g,t}^{\text{TGU}}$ and $F_{g,t}^{\text{TGU}}$ are the active power and the energy consumption of the g th TGU, respectively; η_g^{TGU} is CHPs efficiency; $P_{c,t}^{\text{CHP}}$ and $H_{c,t}^{\text{CHP}}$ are the active power and heat power of the c th CHPs, respectively; ξ_c^{CHP} is the heat-to-electric ratio and η_c^{CHP} is its efficiency; $H_{b,t}^{\text{GB}}$ and $F_{b,t}^{\text{GB}}$ are heat power and the energy consumption of the b th GBs, respectively; η_b^{GB} is GBs efficiency; The active and reactive power output limits of TGU are $[P_g^{\text{TGU}}, \overline{P}_g^{\text{TGU}}]$ and $[Q_g^{\text{TGU}}, \overline{Q}_g^{\text{TGU}}]$, respectively; The active and reactive power output limits of CHPs are

$[\underline{P}_c^{\text{CHP}}, \overline{P}_c^{\text{CHP}}]$ and $[\underline{Q}_c^{\text{CHP}}, \overline{Q}_c^{\text{CHP}}]$, respectively; The heat power limits of CHPs and GBs are $[\underline{H}_c^{\text{CHP}}, \overline{H}_c^{\text{CHP}}]$ and $[\underline{H}_b^{\text{GB}}, \overline{H}_b^{\text{GB}}]$, respectively; The ramp limits of TGU, CHPs and GBs are $\Delta \overline{P}_g^{\text{TGU}}$, $\Delta \overline{P}_c^{\text{CHP}}$ and $\Delta \overline{H}_b^{\text{GB}}$, respectively.

B. Model of Heat Network

The thermal network model involves mass constraint and temperature constraint. The quality constraint is as follows

$$\sum_p m_{p,t} - \sum_q m_{q,t} = m_{n,t}^{\text{in}} - m_{n,t}^{\text{out}} \quad (16)$$

$$\underline{m}_p \leq m_{p,t} \leq \overline{m}_p, \quad \underline{m}_n^{\text{in}} \leq m_{n,t}^{\text{in}} \leq \overline{m}_n^{\text{in}} \quad (17)$$

where $m_{p,t}$ is the mass flow of pipeline p ; $m_{n,t}^{\text{in}}$ and $m_{n,t}^{\text{out}}$ are the inflow and outflow mass at node n , respectively. $\underline{m}_p$ and $\overline{m}_p$ are the lower and upper mass bounds of pipeline p ; $\underline{m}_n^{\text{in}}$ and $\overline{m}_n^{\text{in}}$ are the lower and upper mass bounds at node n .

The temperature constraints are shown as

$$\begin{aligned} H_t^{\text{in}} &= \left(\sum H_{c,t}^{\text{CHP}} + \sum H_{b,t}^{\text{GB}} \right) \\ &= \sum [c^w m_{n,t}^{\text{in}} (T_{n,t}^{\text{s,in}} - T_{n,t}^{\text{r}})] \end{aligned} \quad (18)$$

$$H_t^{\text{out}} = \sum c^w m_{n,t}^{\text{out}} (T_{n,t}^{\text{s}} - T_{n,t}^{\text{r,out}}) \quad (19)$$

$$T_{p,t}^{\text{s,out}} = T_{p,t}^{\text{s,in}}, T_{p,t}^{\text{r,out}} = T_{p,t}^{\text{r,in}}, t^* = t - \tau_{p,t} \quad (20)$$

$$T_{n,t}^{\text{s}} = \frac{T_{n,t}^{\text{s,in}} m_{n,t}^{\text{in}} + \sum T_{p,t}^{\text{s,out}} m_{p,t}}{m_{n,t}^{\text{in}} + \sum m_{p,t}} \quad (21)$$

$$T_{n,t}^{\text{r}} = \frac{T_{n,t}^{\text{r,out}} m_{n,t}^{\text{out}} + \sum T_{p,t}^{\text{r,out}} m_{p,t}}{m_{n,t}^{\text{out}} + \sum m_{p,t}} \quad (22)$$

where H_t^{in} is the total heat power input of heat system; c^w is specific heat capacity; $T_{n,t}^{\text{s,in}}$ is the supply temperature at heat source node n ; $T_{n,t}^{\text{r}}$ is the return temperature at node n ; H_t^{out} is the total heat load; $T_{n,t}^{\text{s}}$ is the temperature of supply pipeline at node n ; $T_{n,t}^{\text{r,out}}$ is the temperature of return pipeline at node n ; $T_{p,t}^{\text{s,in}}$ and $T_{p,t}^{\text{s,out}}$ are input and output temperature of supply pipeline p ; $T_{p,t}^{\text{r,in}}$ and $T_{p,t}^{\text{r,out}}$ are input and output temperature of return pipeline p ; $\tau_{p,t}$ is transmission delay.

C. Model of Electric Network

The power distribution network is modelled as

$$P_{i,t} = \sum P_{g,t}^{\text{TGU}} + \sum P_{c,t}^{\text{CHP}} + \sum P_{w,t}^{\text{WT}} - P_{i,t}^{\text{out}} \quad (23)$$

$$Q_{i,t} = \sum Q_{g,t}^{\text{TGU}} + \sum Q_{c,t}^{\text{CHP}} + \sum Q_{w,t}^{\text{WT}} - Q_{i,t}^{\text{out}} \quad (24)$$

$$\sum_{k:j \rightarrow k} P_{jk} = \sum_{i:i \rightarrow j} P_{ij} - P_j, \underline{P}_{ij} \leq P_{ij,t} \leq \bar{P}_{ij} \quad (25)$$

$$\sum_{k:j \rightarrow k} Q_{jk} = \sum_{i:i \rightarrow j} Q_{ij} - Q_j, \underline{Q}_{ij} \leq Q_{ij,t} \leq \bar{Q}_{ij} \quad (26)$$

$$U_j = U_i - \frac{R_{ij} P_{ij} + X_{ij} Q_{ij}}{U_0}, \underline{U}_j \leq |U_{i,t}| \leq \bar{U}_i \quad (27)$$

where $P_{i,t}$ and $Q_{i,t}$ are the active and reactive injected power of node i , respectively; $P_{i,t}^{\text{out}}$ and $Q_{i,t}^{\text{out}}$ the active and reactive load of node i , respectively; $P_{w,t}^{\text{WT}}$ and $Q_{w,t}^{\text{WT}}$ are the active and reactive power of the w th WT, respectively; P_{ij} and Q_{ij} are the active and reactive power from node i to node j , respectively; $[\underline{P}_{ij}, \bar{P}_{ij}]$ and $[\underline{Q}_{ij}, \bar{Q}_{ij}]$ are limits of P_{ij} and Q_{ij} , respectively; R_{ij} and X_{ij} are the resistance and reactance of branch ij ; U_i is the voltage amplitude at node i ; U_0 is nominal voltage; $[\underline{U}_i, \bar{U}_i]$ is the limit of U_i .

D. Model of Four Control Modes of Heat System

The heating system has different control modes. According to the mass flow rate and heat source temperature of the heating system, it can be divided into four types: constant flow constant heating temperature mode CF-CT, constant flow variable heating temperature mode CF-VT, variable flow constant heating temperature mode VF-CT and variable flow variable heating temperature mode VF-VST. For the three control modes of CF-CT, CF-VT and VF-CT, additional constraints are needed to restrict the control mode of the thermal system, which can be expressed as:

$$m_{n,t}^{\text{in}} = \widetilde{m}_n^{\text{in}}, T_{n,t}^{\text{s,in}} = \widetilde{T}_n^{\text{s,in}}: \text{CF-CT} \quad (28)$$

$$m_{n,t}^{\text{in}} = \widetilde{m}_n^{\text{in}}: \text{CF-VT} \quad (29)$$

$$T_{n,t}^{\text{s,in}} = \widetilde{T}_n^{\text{s,in}}: \text{VF-CT} \quad (30)$$

where $\widetilde{m}_n^{\text{in}}$ and $\widetilde{T}_n^{\text{s,in}}$ indicate the predefined water supply mass flow and heat supply temperature of the system.

E. Objective of Combined Heat and Power Dispatch Model

The goal of the cogeneration model is to minimize the cost of fuel and the penalty cost of wind power abandonment. The formula is

$$\begin{aligned} J = \sum \Delta t \left\{ \left[\sum F_{g,t}^{\text{TGU}} + \sum F_{c,t}^{\text{CHP}} + \sum F_{b,t}^{\text{GB}} \right] \lambda^{\text{gas}} \right. \\ \left. + \sum (\overline{P}_{w,t}^{\text{WT}} - P_{w,t}^{\text{WT}}) \lambda^{\text{WT}} \right\} \end{aligned} \quad (31)$$

where $\overline{P}_{w,t}^{\text{WT}}$ is the prediction power of the w th WT; λ^{gas} and λ^{WT} are the cost of fuel and curtailment.

IV. CASE STUDY

In this case, the integrated electric-thermal energy system consists of an IEEE 33-bus distribution network and a 6-node district heating network. The system diagram is shown in Figure 1.

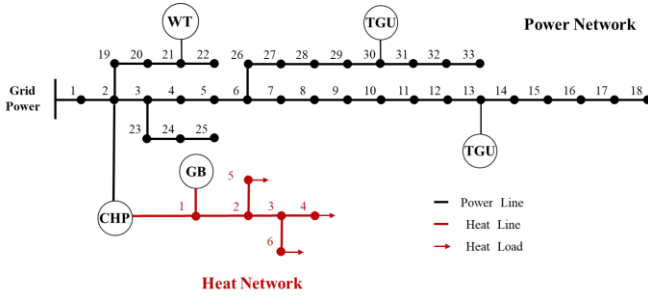


Fig. 1. Diagram of the integrated electric-thermal energy system.

The wind power data of a wind farm in northern China are used to obtain the probability prediction results, as shown in Figure 2. The confidence level of the forecast interval was set at 90%.

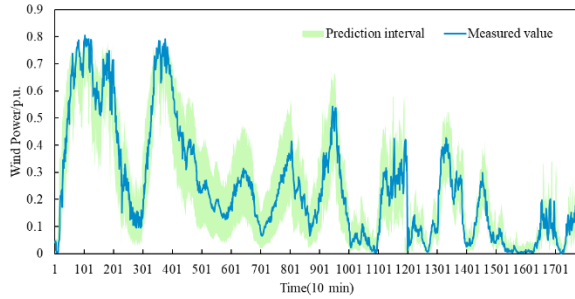


Fig. 2. Probabilistic prediction result of wind power

In order to evaluate the quality of probabilistic prediction results obtained by ELM method, the prediction interval coverage probability (PICP), average coverage error (ACE), average width of prediction interval (PIAW) and Winkler interval score (S_w) were used. The first two indicators reflect the reliability of probabilistic prediction, and the last two indicators reflect the sharpness of the method. Table 1 lists the values of these metrics.

TABLE I. EVALUATION RESULT

Indice	PICP	ACD	AW	S_w
Value	91.07%	1.07%	0.2538	0.2976

It can be seen from Table 1 that the ELM probability prediction method can obtain relatively sharp results and has high prediction reliability. Considering the uncertainty of wind power generation, the expectation of ELM probability is used as the input of combined heat and power dispatching. Some parameters of the electric heating integration system are shown in Table 2.

TABLE II. SYSTEM PARAMETERS

Parameter	Value	Parameter	Value
$[P_g^{TGU}, \bar{P}_g^{TGU}]$	[30,50]MW	$\Delta \bar{P}_g^{TGU}$	10MW
$[H_c^{CHP}, \bar{H}_c^{CHP}]$	[20,50]MW	$\Delta \bar{P}_c^{CHP}$	10MW
$[H_b^{GB}, \bar{H}_b^{GB}]$	[15,50]MW	$\Delta \bar{H}_b^{GB}$	10MW
$[\underline{m}_n^{in}, \bar{m}_n^{in}]$	[80,800]kg/s	η_g^{TGU}	0.25
η_c^{CHP}	0.72	η_b^{GB}	0.93

The dispatch results of different control modes are shown in the Table 3. Due to the non-flexible property of CF-CT model, the feasible solution cannot be derived by commercial solving tools. The comparison results show that, among the other three control modes, the VF-VT mode has the smallest wind power reduction and the optimal total cost, which indicates the optimality of the control mode.

TABLE III. COMBINED HEAT AND POWER DISPATCH RESULT

Control Mode	CF-CT	CF-VT	VF-CT	VF-VT
Total Cost ($\times 10^5 \$$)	/	3.21	3.65	2.47
Curtailment (MWh)	/	6.37	6.58	5.23

V. CONCLUSION

Considering the uncertainty of wind power generation, a method of cogeneration under different control modes of thermal systems is proposed. The ELM method is used as the probabilistic prediction model, and the prediction results are set as the input of the combined heat and power scheduling model under different control modes. Taking the electro-thermal integrated energy system as an example, the research results show that VF-VT mode has the best performance among the four different control modes considering the uncertainty of wind power prediction.

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